Differential Geometry

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1 Manifolds

The concept of a manifold is intended to generalise curves and surfaces in $\mathbb{R}^3$.

Definition 1.1 (smooth structure)

A *smooth structure* on a topological space M is a collection of *charts* (an *atlas*)

$$\varphi_\alpha : U_\alpha \to V_\alpha$$

where $U_\alpha \subseteq M$, $V_\alpha \subseteq \mathbb{R}^d$ are open, φ_α is a homeomorphism, such that

1. $M = \bigcup_\alpha U_\alpha$. That is, the charts cover M.
2. Given any charts $\varphi_\alpha, \varphi_\beta$, the map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a smooth map between open subsets of $\mathbb{R}^d$.
3. If φ is *compatible* (as in the previous point) with every chart in the atlas, then φ is a chart in the atlas.

In practice (i.e. using Zorn's lemma), given any collection of charts satisfying the first two conditions, we can find a maximal atlas containing it.

Remark 1.2.
1. The second condition implies that $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a diffeomorphism.
2. d is fixed, assuming that M is connected.

Definition 1.3 (manifold)

A *(smooth) manifold* M is a second countable Huasdorff space topological space, equipped with a smooth structure.

$d = \dim(M)$ is the *dimension* of M.

In practice, we may induce a topology on M from a smooth structure (without the mention of open or homeomorphism), then $D \subseteq M$ is open if for any chart φ_α, $\varphi_\alpha(D \cap U_\alpha)$ is an open subseteq of $\mathbb{R}^d$.

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Remark 1.4. 1. We can also define a C^k manifold by replacing smooth with C^k as above. If $k = 0$, then we have a *topological manifold*.

2. On the other hand, we can replace $\mathbb{R}^d$ with $\mathbb{C}^n$, and obtain a *complex manifold*.

$\mathbb{R}^n$ is (trivially) a manifold.

Example 1.5 (unit sphere)

Define

$$S^n = \{x = (x_0, \dots, x_n) \in \mathbb{R}^n \mid ||x|| = 1\} \subseteq \mathbb{R}^{n+1}$$

For the charts, we can consider the stereographic projections

$$\varphi(x) = \frac{1}{1 - x_0}(x_1, \dots, x_n) \qquad \text{for } x_0 \neq 1$$

and

$$\psi(x) = \frac{1}{1 + x_0}(x_1, \dots, x_n) \qquad \text{for } x_0 \neq -1$$

In particular,

$$\psi \circ \varphi^{-1}(u) = \frac{u}{||u||^2}$$

Example 1.6

If M, N are manifolds of dimensions d, e respectively, then $M \times N$ is a manifold of dimension $d + e$.

As a corollary, $T^n = (S^1)^n$ is a manifold.

Example 1.7

If M is a manifold, $X \subseteq M$ is open, then X is a manifold.

Lecture 2

A convention, we will will assume for any manifold M, we will assume all components are of the same dimension. In particular, $\dim(M)$ is well defined.

Notation 1.8. We will write M^d to say M is a manifold of dimension d.

Example 1.9 (real projective space)

The real projective space is

$$\mathbb{RP}^n = \{\text{straight lines in } \mathbb{R}^{n+1} \text{ through } 0\}$$

Points in $\mathbb{RP}^n$ can be written using homogeneous coordinates $(x_0 : \cdots : x_n)$, where the x_i are not all zero, and

$$(x_0 : \cdots : x_n) = (\lambda x_0 : \cdots : \lambda x_n)$$

for all $\lambda \neq 0$. We will induce a topology from a smooth structure. The charts are given by (φ_i, U_i) where

$$U_i = \{x_i \neq 0\}$$

and we define

$$\varphi_i(x_0 : \cdots : x_n) = \left(\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right) \in \mathbb{R}^n$$

For $i < j$,

$$\varphi_j \circ \varphi_i^{-1}(y_0, \dots, y_n) = \left(\frac{y_1}{y_j}, \dots, \frac{1}{y_j}, \dots, \frac{y_n}{y_j} \right)$$

where the 1 is in the i-th position, and we omit the j-th position. Thus, $\mathbb{RP}^n$ is an n-dimensional manifold. Similarly, we can define the complex projective space $\mathbb{CP}^n$, which is a $2n$-manifold (and a n-dimensional complex manifold).

Example 1.10 (Grassmannian)

$$\mathrm{Gr}(k, n) = \{k\text{-dimensional subspaces of } \mathbb{R}^n\}$$

Then $\mathrm{Gr}(k, n)$ is a manifold of dimension $k(n-k)$. Note we can also define the Grassmannian over $\mathbb{C}$.

Sketch proof. An example of a coordinate neighbourhood is

$$U = \left\{k\text{-dimensional subspaces obtainable as the span of rows of } k \times n \text{ matrices of the form } \begin{pmatrix} I_k & * \end{pmatrix}\right\}$$

where I_k is the $k \times k$ identity matrix. In this case, the $k \times (n-k)$ block $*$ defines local coordinates. Choosing different columns, we have $\binom{n}{k}$ neighbourhoods, $U_{i_1,\dots,i_k}$, where $i_1 < i_2 < \cdots < i_k$. □

In particular, note that $\mathbb{RP}^n = \mathrm{Gr}(1, n+1)$.

Example 1.11 (Non-example)

Consider an equivalence relation on $\mathbb{R}^2$ given by

$$(x, y) \sim \left(\lambda x, \frac{1}{\lambda} y \right)$$

for any $\lambda \neq 0$. Let $X = \mathbb{R}^2/\sim$ for the quotient, with the quotient topology. We can consider

$$\{(x, y) \mid xy = c\}$$

If $c \neq 0$, then this is one equivalence class. If $c = 0$, this splits into three, which are

- $\{(0, 0)\}$,
- $\{(x, 0) \mid x \neq 0\}$,
- $\{(0, y) \mid y \neq 0\}$

We can write

$$X = (-\infty, 0) \cup (0, \infty) \cup \{0', 0'', 0'''\}$$

We have three charts φ_i corresponding to each choice of zero. For example,

$$U_1 = (-\infty, 0) \cup \{0'\} \cup (0, \infty)$$

with the "natural" map φ_1. On the other hand, the induced topology is not Hausdorff. Note the induced topology is *not* the quotient topology.

Refer to Examples Sheet 1 Question 12 for an example of a non-second countable (non-)example.

Definition 1.12 (smooth map)

Let M, N be manifolds, a continuous map $f : M \to N$ is *smooth* if for any $p \in M$, we can find charts (φ, U) near p, (ψ, V) near $f(p)$, and the composition

$$\psi \circ f \circ \varphi^{-1}$$

is smooth. as a map on

$$\varphi(U \cap f^{-1}(V))$$

Definition 1.13 (diffeomorphism)

A smooth map $f : M \to N$ is a *diffeomorphism* if f is a bijection, and f^{-1} is smooth. In this case, we say that M, N are *diffeomorphic*.

Proposition 1.14.

1. If $U \subseteq \mathbb{R}^n$, then $f : U \to \mathbb{R}^m$ is smooth in the above definition if and only if it is smooth in the sense of multi-variate calculus.
2. Every chart $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^d$ is a smooth map of manifolds. Moreover, it is a diffeomorphism.
3. If $f : M \to N$, $g : N \to P$ are smooth, then $g \circ f : M \to P$ is smooth.

Proof. Obvious, and so omitted. □

1.1 Matrix Lie groups

Consider the group $\mathrm{GL}(n, \mathbb{R}) \subseteq \mathbb{R}^{n^2}$. In fact, this is an open subset, and so it is a manifold. Moreover, matrix multiplication and inversion are smooth.

Definition 1.15 (Lie group)

A group G is called a *Lie group* if

- G is a manifold,
- the multiplication map $G \times G \to G$ is smooth,
- inversion $G \to G$ is smooth.

Remark 1.16. Suffices to show that the map

$$G \times G \ni (\sigma, \tau) \mapsto \sigma\tau^{-1} \in G$$

is smooth.

In particular, $\mathrm{GL}(n, \mathbb{R}$ is a Lie group. Similarly, $\mathrm{GL}(n, \mathbb{C})$ is also a Lie group. More generally, for $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{C})$, we define the norm $\|A\| = n \max_{i,j} |i, j|$. (Note in finite dimensional, all norms are equivalent). In particular,

$$\|AB\| \leq \|A\|\|B\|$$

We define the *exponential of* A as

$$\exp(A) = \sum_k \frac{A^k}{k!}$$

This series converges for all A, by the same proof as for $\exp : \mathbb{C} \to \mathbb{C}$ (or more generally, any Banach algebra). Moreover, it converges uniformly on balls

$$\{\|A\| \leq R\}$$

by the Weierstrass M-test. In particular, $\exp : \mathrm{Mat}_n(\mathbb{C}) \to \mathrm{Mat}_n(\mathbb{C})$ is continuous.

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